

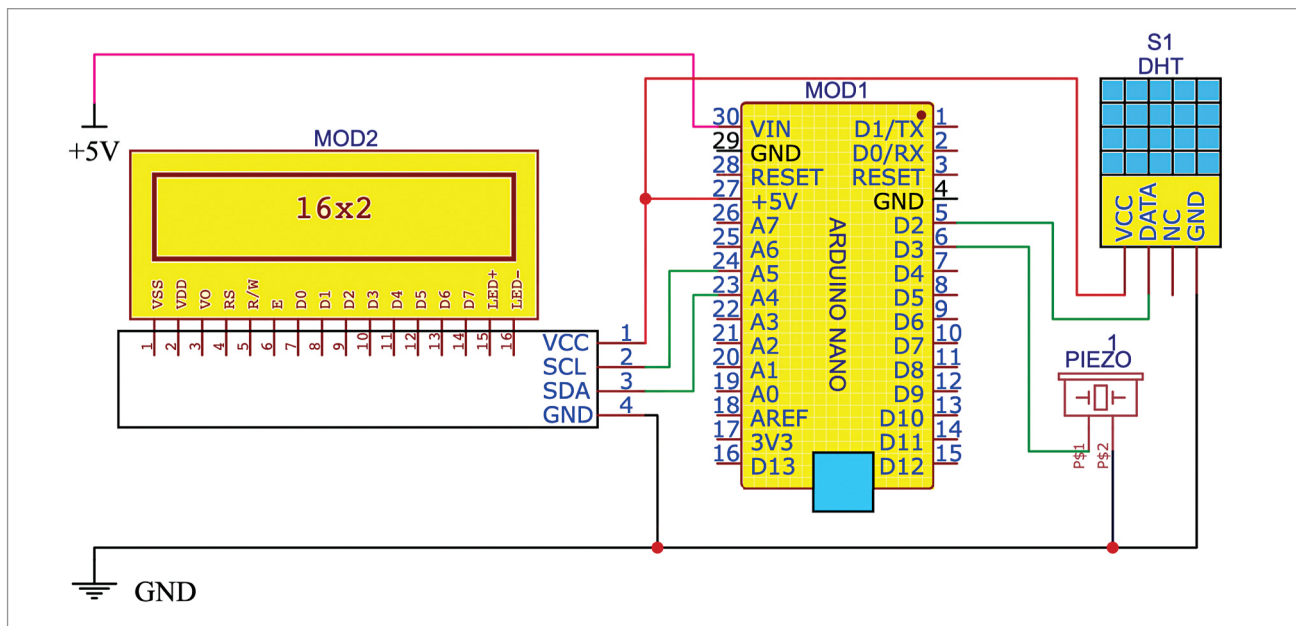


Fig. 2: Circuit diagram

A standard I2C LCD display (MOD2) consists of an HD44780-based character LCD display coupled with an I2C LCD adaptor. These displays are adept at presenting character-based information. For instance, a 16×2 character LCD can accommodate 32 ASCII characters across two rows. At the core of the adaptor lies an 8-bit I/O expander chip, the PCF8574, which converts Arduino's I2C data into the parallel data necessary for the LCD display.

Software aspects

Programming for this project was accomplished using Arduino 1.8.19 tool. The DHT11 sensor's input is directed to Arduino Nano's pin D2, allowing the display of temperature and humidity readings both for the surroundings and the field on the LCD module. An attached piezo buzzer, connected to pin D3, serves as an alarm system to signal excessive temperatures. The loop function encompasses the data checking and alarm generation process.

First write the source code and save the sketch (software) in .ino

format. Before uploading the sketch program to the Arduino Nano board, ensure the correct board is selected. Following board selection, proceed with uploading the source code. Source code snippet for the loop function is exhibited in Fig. 3.

Construction and testing

Upon uploading the source code to the Arduino Nano, assemble the project either on a breadboard or a general-purpose PCB. After completing the circuit connections, enclose the assembly in a suitable housing, incorporating the Arduino Nano board. Verify connections according to the circuit diagram and supply 5V DC voltage to the assembled circuit. Monitor temperature and humidity status on the LCD screen. If the

```
void loop() {
  float temp = dht.getTemperature();
  float humi = dht.getHumidity();
  if (temp > 25)
  {
    digitalWrite(buzzerPin,HIGH);
  }
  lcd.setCursor(0,0);
  lcd.print("Temp: ");
  lcd.print(temp);
  lcd.print(" C");
  lcd.setCursor(0,1);
  lcd.print("Humi: ");
  lcd.print(humi);
  lcd.print(" %");
  delay(2000);
}
```

Fig. 3: Source code snippet of loop function

temperature approaches the critical threshold of 25°C , the integrated piezo buzzer will sound, prompting farmers to take necessary measures based on weather and climatic conditions. **EFY**

EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com



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